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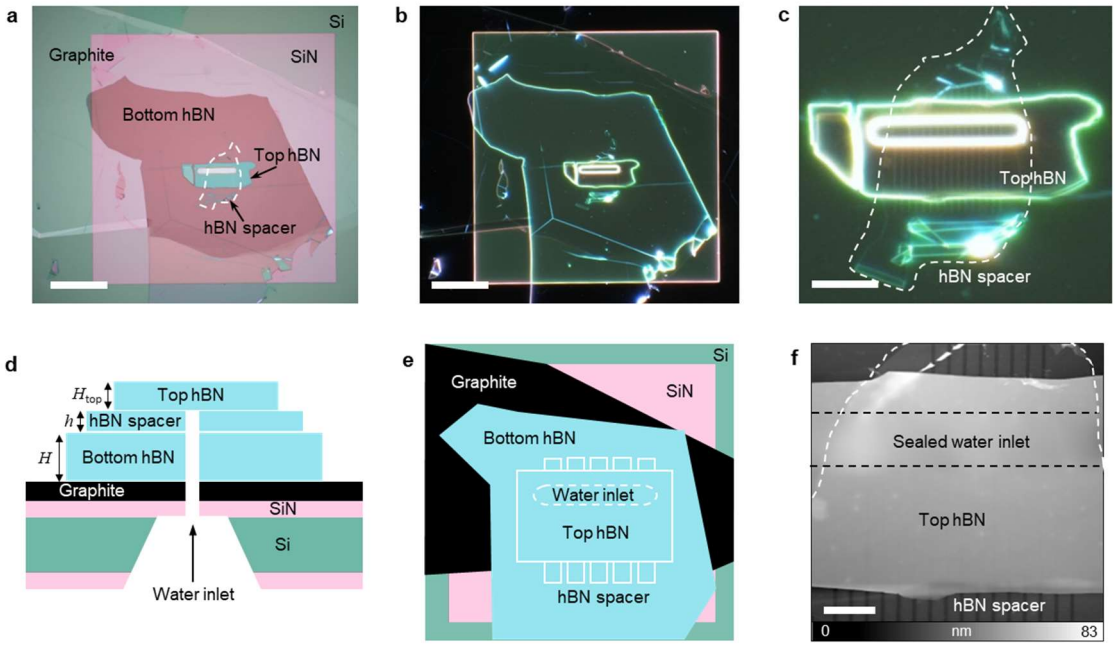
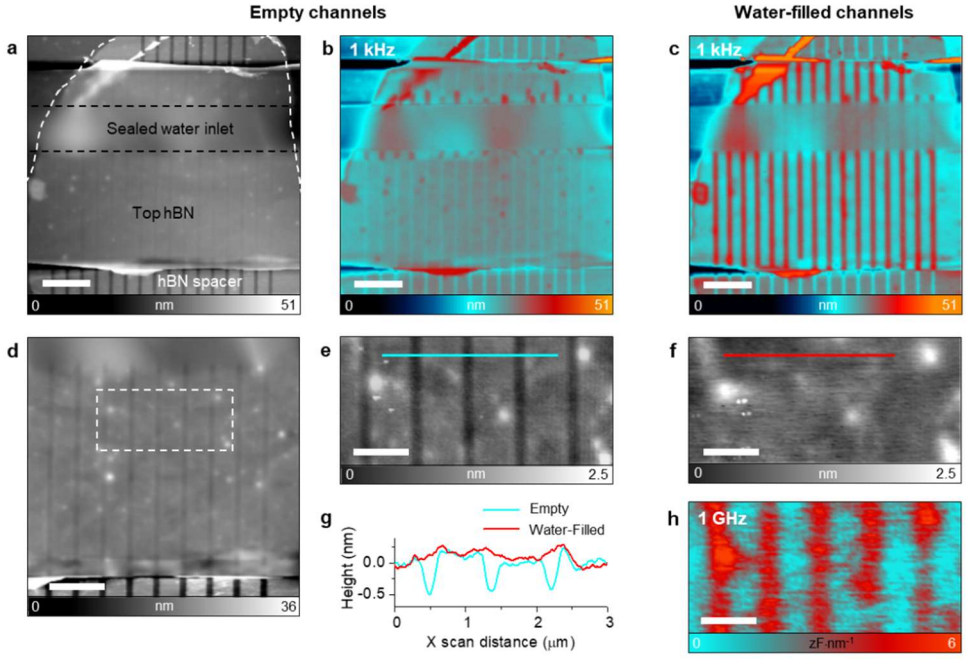


Extended Data Fig. 1| Devices for in-plane dielectric imaging of confined water. (a) Optical image of a representative device with the channel height $h \approx 5$ nm. The edges of the spacer layer are indicated by the white dashed lines. The large square region (appears in pink) is the suspended SiN membrane. It has a ~ 20 μm-long rectangular slit in its center (white) that connects nanochannels to a water reservoir at the backside of the device. **(b)** Dark-field optical image of the same device and **(c)** Zoom into its central region where numerous parallel channels can be seen**. (d**,**e)** Sketch of the device geometry. Its cross-section (d) and top-view (e) (not to scale). **(f)** AFM topography image of the device's central region. The black dashed lines indicate the aperture's position. The white dashed curve indicates edges of the hBN spacer crystal. Scalebars: (a,b) 20 μm; (c) 10 μm; and (f) 3 μm.



Extended Data Fig. 2| Filling nanochannels with water. (a) AFM topography image of a representative device with $h \approx 5$ nm before filling it with water – same image as in Extended Data Fig. 1f but using a flattening filter to better visualize the sagging of the top hBN layer into nanochannels. The image was taken in the intermittent-contact attractive mode with no voltage applied. **(b,c)** Same as (a) but with applying a low-f electric field (v_{AC} = 8V, 1 kHz) before (b) and after (c) filling the nanochannels with water. The contrast associated with the channels changes from light cyan in (b) to red in (c), indicating that water filled all the channels. **(d)** Zoom-in of the central region in (a). The top hBN layer is partially sagged into the empty channels. **(e)** Further zoom-in showing the region indicated by the dashed rectangular in (d). **(f)** Same as (e) but after filling nanochannels with water. It shows that the top hBN layer became flat, no longer sagging inside the nanochannels. **(g)** Topography profiles over three channels in (e,f), as indicated by color-coded lines. **(h)** Dielectric image taken from the same region as in (f) at a constant height and at 1 GHz. Scalebars: (a-c) 3 μm, (d) 2 μm, (e,f,h) 1 μm.